Bayesian statistics with R
9. Conclusions

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November-December 2026

Remember the objectives

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- Try and demystify Bayesian statistics, and what we call MCMC.
- Make the difference between Bayesian and Frequentist analyses.
- Understand the Methods section of ecological papers doing Bayesian stuff.
- Run Bayesian analyses, safely hopefully.

What was on our plate?

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1. An introduction to Bayesian inference
2. The likelihood
3. Bayesian analyses by hand
4. A detour to explore priors
5. Markov chains Monte Carlo methods (MCMC)
6. Bayesian analyses in `R` with the `Jags` software
7. Contrast scientific hypotheses with model selection
8. Heterogeneity and multilevel models (aka mixed models)
9. Conclusions

Free the modeler in you (M. Kéry)

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- Uses probability to quantify uncertainty for everything (propagation of uncertainty).
- Allows use of prior information ('better' estimates: avoid boundary param estimates, acknowledge existing knowledge).
- Can fit complex (hierarchical) models with same MCMC algorithms.
- Credible intervals make more sense than confidence intervals

With great tools come great responsabilities

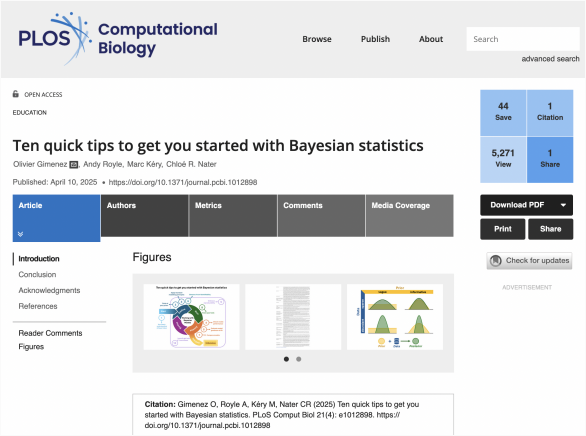
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- Checking convergence is painful.
- Specifying priors might be tricky.
- Model adequacy should be checked (posterior predictive checks - not covered).
- Computational burden can be high (see function `R2jags::jags.parallel()` and package 'jagsUI'.

Tips and tricks

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https://hal.science/CEFE/hal-04731240v2



So what?

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- Make an informed and pragmatic choice.
- Are you after complexity, speed, uncertainties, etc?
- Talk to colleagues.